

Investigating Antibacterial Effects of Garlic (*Allium sativum*) Concentrate and Garlic-Derived Organosulfur Compounds on *Campylobacter jejuni* by Using Fourier Transform Infrared Spectroscopy, Raman Spectroscopy, and Electron Microscopy

Fourier transform infrared (FT-IR) spectroscopy and Raman spectroscopy were used to study the cell injury and inactivation of *Campylobacter jejuni* from exposure to antioxidants from garlic. *C. jejuni* was treated with various concentrations of garlic concentrate and garlic-derived organosulfur compounds in growth media and saline at 4, 22, and 35°C. The antimicrobial activities of the diallyl sulfides increased with the number of sulfur atoms (diallyl sulfide < diallyl disulfide < diallyl trisulfide). FT-IR spectroscopy confirmed that organosulfur compounds are responsible for the substantial antimicrobial activity of garlic, much greater than those of garlic phenolic compounds, as indicated by changes in the spectral features of proteins, lipids, and polysaccharides in the bacterial cell membranes. Confocal Raman microscopy (532-nm-gold-particle substrate) and Raman mapping of a single bacterium confirmed the intracellular uptake of sulfur and phenolic components. Scanning electron microscopy (SEM) and transmission electron microscopy (TEM) were employed to verify cell damage. Principal-component analysis (PCA), discriminant function analysis (DFA), and soft independent modeling of class analogs (SIMCA) were performed, and results were cross validated to differentiate bacteria based upon the degree of cell injury. Partial least-squares regression (PLSR) was employed to quantify and predict actual numbers of healthy and injured bacterial cells remaining following treatment. PLSR-based loading plots were investigated to further verify the changes in the cell membrane of *C. jejuni* treated with organosulfur compounds. We demonstrated that bacterial injury and inactivation could be accurately investigated by complementary infrared and Raman spectroscopies using a chemical-based, “whole-organism fingerprint” with the aid of chemometrics and electron microscopy.